

Sebastian Villalobos

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EDUCATION

GEORGETOWN UNIVERSITY

Washington, DC

M.S. in Data Science & Analytics

Expected May 2027 | GPA: 4.0/4.0

- Teaching Assistant: Database Systems and SQL course

UNIVERSIDAD IBEROAMERICANA

Mexico City, MX

Bachelor's degree in Mechatronics and Production Engineering

- Honors: Awarded Tuition Scholarship for academic merit

WORK EXPERIENCE

WELO DATA

Mexico City, MX

Data Science & Analytics Engineer

Nov 2024 – Aug 2025

- Developed XGBoost model predicting task difficulty across 120k+ records, improving annotation accuracy by 10% via optimized task allocation
- Developed GMM-based recommendation system for rater-project matching, improving annotation quality by 15% and 10% across two projects
- Automated financial ETL pipeline in Redshift processing high-volume transactions (0.001% error rate), reducing reconciliation time by 70%
- Led analytics for two projects, building end-to-end ML pipelines for real-time KPI monitoring

SCALE AI

Mexico City, MX

Data Operations Engineer

Oct 2022 – Nov 2024

- Developed automated evaluation frameworks for NVIDIA Nemotron's RLHF alignment using GPT-4 as Judge-LLM and HelpSteer2 methodology
- Built LLM quality linting system adopted across GenAI organization, automating error detection via programmatic flagging and reducing manual review by 40%
- Built TypeScript-MongoDB integrations and Python pipelines to automate end-to-end customer data processing, eliminating manual bottlenecks

Operations Data Analyst & Specialist

Feb 2021 – Oct 2022

- Built Logistic Regression model on 12+ months of operational data to predict batch acceptance, improving approval rates by 20%

PROJECTS

ModelLens — NN Interpretability Toolkit ([GitHub](#))

Georgetown University - In Progress

- Built open-source PyTorch interpretability package supporting activation patching, logit lens, residual stream analysis, attention visualization, and clean/corrupted run comparisons across HuggingFace and vanilla PyTorch backends
- Designed adapter architecture enabling unified analysis across HuggingFace and vanilla PyTorch models with automatic backend detection

Stoic LLM Steering ([GitHub](#))

Personal Project - In Progress

- Implementing LLM behavioral control using Contrastive Activation Analysis (CAA) and LoRA fine-tuning, extracting steering vectors from Stoic vs. non-Stoic text pairs for inference-time style adjustment

Political Sentiment Analysis ([GitHub](#))

Georgetown University

- Developed DistilBERT-based NLP pipeline for emotion classification across 2,000+ Congressional speeches, identifying statistically significant partisan patterns

PUBLICATIONS

Villalobos-Alva, S., et al. (2022). Protein Science Meets AI: Systematic Review. Front. Bioeng. Biotechnol. [\[Link\]](#)

SKILLS & OTHER INFORMATION

AI & Machine Learning: Generative AI (LLMs, RLHF, HelpSteer2), Mechanistic Interpretability (Activation Patching, Logit Lens, Residual Stream Analysis), NLP/NLU, Computer Vision, Deep Learning, LLM Evaluation (Judge-LLM), XGBoost, GMM
Programming: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face), SQL, NoSQL, TypeScript, R, Git, GitHub, Linux/CLI

Languages & Certifications: Spanish (Native), English (Fluent); MIT MicroMasters in Statistics and Data Science